Vanguard User Manual

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1.0	March 2, 2008	Joanne Muldoon/ Christine Danner/Janet Miler	Major	New Release (formally known as Operators Guide Part 1-1-0102A)

Vanguard User Manual

Vanguard Breast MRI Auxiliary Table® with 8 Channel Coil Array for GE Signa™ Systems

(For sale in Canada and the United States of America only)

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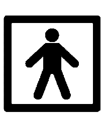

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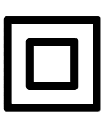

Class II Equipment

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Chapter 1 Introduction

User Manual Information

This manual describes the safety precautions, function, features and methods of use and care for the Vanguard Breast MRI Auxiliary Table with 8-Channel Coil Array for GE Signa™ 1.5T, to be referred to as the Vanguard herein. Please review this manual entirely before using the system.

Use of Symbols

Table 1-1

Safety Notice	Meaning
DANGER	A red **DANGER** symbol is used to identify conditions or actions for which a specific hazard is known to exist. These condition or actions will cause severe personal injury, death or substantial property damage if the instructions are not followed.
WARNING	An orange **WARNING** symbol is used to identify conditions or actions for which a specific hazard is known to exist. These conditions or actions may cause severe personal injury or substantial property damage if the instructions are not followed.
CAUTION	A yellow **CAUTION** symbol is used to identify conditions or actions for which a potential hazard may exist. These conditions or actions may cause minor personal injury or property damage if the instructions are not followed.
CAUTION	Do not cross or loop cables; arcing and patient burns can result.

Safety Notice	Meaning
NOTE	A **Note** provides additional information that is helpful to the user. It may emphasize certain information regarding special tools or techniques, items to check before proceeding, or factors to consider about a concept or task.
 **NO PINCH**	Do not place body parts or cables in areas marked with this label as they will be pinched and could cause harm or damage.
 **WARNING**	Ferromagnetic components caution. Do not use magnetic or magnetically attractive items in or around these labelled areas.

Safety

The following general warning statements apply to scanning with this system in a magnetic resonance (MR) scanner. For further details, review the warnings included in your MR system operator's guide.

Table 1-2

Safety Notice	Meaning
 **WARNING**	Ensure that the patient does not form a loop with any body parts. Do not allow patients to touch any part of the right hand or arm to any part of the left hand or arm. The loop formed by doing so could cause an RF burn at the point of contact. Use pads to ensure the patient's hands do not touch any metal features of the Vanguard. Be sure to educate the patient accordingly and check the patient's position immediately before the scan.
 **WARNING**	This coil system is compatible only with GE Signa™ 1.5T Scanners.
	Do not use coils, cables, or gating leads with exposed metal surfaces or abraded insulation.
	Remove unused or improperly connected coils, cables, and gating leads from the MR scanner before performing a scan.
	Check that the cable or lead(s) does not touch the sides of the MR scanner.

Safety Notice	Meaning
	Immediately stop the MR scan acquisition if the patient complains of a tingling and/or heating sensation.
	All personnel using the system should be instructed in the proper connection and handling of the Vanguard and should be familiar with this manual.
	Auxiliary equipment (such as gating equipment, vital sign monitoring systems and RF coils) that has not been specifically tested and approved for use in the MR environment may result in burns or other injuries to the patient. This equipment may also interfere with the proper operation of this system.
	The coil is not suitable for use in the presence of flammable anaesthetics in combination with air, oxygen or nitrous oxide.
	Service personnel must have specialized training to ensure the safe operating condition of the Vanguard Therefore, only properly trained and qualified personnel should be authorized to service any components.
	Before using the Vanguard, visually check it to ensure there is no external damage. Do not use a coil if the housing or cable is broken.
 CAUTION	Ensure correct connection of the coil cables as described in this manual.
	Position the patient, coils and the cables as prescribed in this manual.
	Check that the coil cable does not directly contact any part of the patient.
	Any modifications made to the Vanguard that are not authorized by Sentinelle Medical Inc. may void the warranty. Any changes made may impact the safety of the Vanguard.
	Remove any accessory padding from or around the bore before the patient support advances into the MR scanner.

Safety Notice	Meaning
 CAUTION	Do not cross or loop cables; arcing and patient burns can result.
 NO PINCH	Ensure cables or body parts are not pinched during patient support travel into or out of the magnet.
NO STEP	To avoid injury do not step on the stretcher drawers or stretcher base.

Emergency Procedures

Emergency Undocking

If the Vanguard fails to undock using the undock pedal, it can be detached from the MR scanner by pulling the RED **Emergency Undock** lever at the base of the GE Signa™ scanner (see GE Signa™ User's Manual for further details).


CAUTION

In order to appropriately use the Emergency Undock lever at the base of the GE Signa™ scanner the patient support must be in the home position.

Emergency Release

Patient removal with the emergency release handle (Figure 1) can be used in the event of a power outage in the magnet room, after pressing the **Emergency Stop** button, or in case of a patient emergency. Ensure the bridge is in the locked position when performing an emergency removal.

The handle at the head end of the patient support is the emergency release. The emergency release is activated by manually twisting the handle toward you and pulling the patient support out of the MR scanner.

Figure 1: Emergency Release Handle

Note: Be sure to disconnect the Vanguard cable prior to using the emergency release handle.

Once the patient support is in home position, the stretcher can be undocked from the MR scanner by depressing the **UNDOCK** pedal.

Reporting of Incidents

Please contact Sentinelle Medical Inc. immediately to report an incident and/or injury to a patient, operator, or maintenance employee that occurred as a result of operation of the Vanguard.

If an accident occurs as a result of operation of the Vanguard, do not operate the equipment until an authorized investigation has been conducted.

Call Sentinelle Medical Inc. 24-hour assistance at: **866-735-3744**

Email specific information to: **feedback@sentinellemedical.com**

Indications for Use/Contra-indications

Indications for Use:

The Vanguard is designed to provide MR images of breast anatomy when used in conjunction with a MR Scanner. These images can be interpreted by a trained physician. When used with the disposable sterile biopsy grid the Vanguard permits access to the breast anatomy for biopsy and localization procedures that can be performed by a trained physician.

Contraindications:

The operator should be aware of the following contraindications for use related to the strong magnetic field of the MR scanner:

Table 1-3

Safety Notice	Meaning
	Scanning is contraindicated for patients who have electrically, magnetically or mechanically activated implants such as cardiac pacemakers. The magnetic and electromagnetic fields produced by the MR scanner and coil may interfere with the operations of these devices.
	Scanning patients with intracranial aneurysm clips is contraindicated.

Coil Description

The Vanguard allows coils to be interchanged and reconfigured as appropriate for a particular imaging task. The coil is positioned close to the patient's breast during imaging. These coils consist of both an 8 different channel configuration arranged in a phased array design and an interchangeable 4 channel and 2 channel for interventional work. The coils are designed to give improved signal-to-noise over that of a standard body coil.

Coil Configurations

The Vanguard offers various coil configurations. When selecting a particular coil configuration, the physical coil arrangement must be consistent with the selected coil configuration. Table 1-4 describes the various coil arrangements, their appropriate application, corresponding coil configuration and corresponding coil.

Table 1-4: Coil Configuration

Coil Configuration	Clinical Application	Coil Selection MRI Console/Physical Setup
8 BrstBL SMI	Used for **Bilateral** imaging applications. In this coil configuration, all eight channels are used. Four channels will be used to image the right breast and four to image the left breast.	MRI CONSOLE: To select this coil mode, first select the coil with description **8ch Bilat HD Breast Array by Sentinelle** on the left side of the coil selection menu on the operator console. Then select the coil mode with description **8 BrstBL SMI** on the right side of the menu. PHYSICAL COIL ARRANGEMENT: To use this imaging mode, the coils denoted **Lateral Array Coil Right VG**, **Lateral Array Coil Left VG** and **Medial Array Coil VG**, must be used. The coils must always be positioned so the label reading, **THIS SIDE TOWARD PATIENT**, is directed towards the patient. All of these coils should be attached to the appropriate connector.

Coil Configuration	**Clinical Application**	**Coil Selection** **MRI Console/Physical Setup**
8 BrstLt SMI	Used for **Unilateral** imaging applications of the left breast.	MRI CONSOLE: To select this coil mode, first select the coil with description **8ch Bilat HD Breast Array by Sentinelle** on the left side of the coil selection menu at the operator console. Then select the coil mode with description **8 BrstLt SMI** on the right side of the menu. PHYSICAL COIL ARRANGEMENT: To use this imaging mode, the coils denoted **Lateral Array Coil Right VG**, **Lateral Array Coil Left VG** and **Medial Array Coil VG**, must be used. The coils must always be positioned so the label, **THIS SIDE TOWARD PATIENT**, is directed towards the patient. All of these coils should be attached to the appropriate connector.
8 BrstRt SMI	Used for **Unilateral** imaging applications of the RIGHT breast.	MRI CONSOLE: To select this coil mode, first select the coil with description **8ch Bilat HD Breast Array by Sentinelle** on the left side of the coil selection menu on the operator console. Then select the coil mode with description **8BrstRt SMI** on the side of the menu. PHYSICAL COIL ARRANGEMENT: To use this imaging mode, the coils denoted **Lateral Array Coil Right VG**, **Lateral Array Coil Left** and **Medial Array Coil VG must be used**. The coils must always be positioned so the label, **THIS SIDE TOWARD**

Coil Configuration	Clinical Application	Coil Selection MRI Console/Physical Setup
8 BrstRt SMI		**PATIENT**, is directed towards the patient. All of these coils should be attached to the appropriate connector.
2 BrstBL int SMI	Used for **Interventional** applications. Used for unilateral breast biopsies for the left and right side.	MRI CONSOLE: To select this coil mode, first select the coil with description **2ch Unilat Biopsy HD Breast Array by Sentinelle** on the left side of the coil selection menu on the operator console. Then select the coil mode with description **2 BrstBL int SMI** on the right side of the menu. PHYSICAL COIL ARRANGEMENT: To use this imaging mode, the coils denoted **Lateral Single Loop Coil Left VG** and **Medial Single Loop Coil Right VG** must be used for a Left breast biopsy. **Lateral Single Loop Coil Right VG** and **Medial Single Loop Coil Left VG** must be used for a right breast biopsy. *The coils must always be positioned so the label,* ***THIS SIDE TOWARD PATIENT****, is directed towards the patient. In both configurations one coil must be placed on the lateral side, and one coil on the medial side of the breast. In both configurations the* ***Medial Plug****, must be plugged into the middle connector on the system.*

Coil Configuration	Clinical Application	Coil Selection MRI Console/Physical Setup
4 BrstBL int SMI	Used for **Bilateral Interventional** applications.	MRI CONSOLE: To select this coil mode, first select the coil with description **4ch Bilat Biopsy HD Breast Array by Sentinelle** on the left side of the coil selection menu on the operator console. Then select the coil mode with description **4 BrstBL int SMI** on the right side of the menu. PHYSICAL COIL ARRANGEMENT: To use this imaging mode, the coils denoted **Lateral Single Loop Coil Right VG** and **Lateral Single Loop Coil Left VG**, should be used with the **Medial Array Coil VG**. The coils must always be positioned so the label, **THIS SIDE TOWARD PATIENT**, is directed towards the patient. Two single loop coils must be used; one on the lateral side of each breast. For bilateral biopsies, an additional set of immobilization frames are used on the medial side of each breast.

Coil Nomenclature

For consistency and ease of use, the labels of the coils will not be used to describe their use. Please refer to Table 1-5 for a listing of these names.

Table 1-5: Coil Name Cross-reference

Coil Label Name	Coil Name Referenced in Manual
Lateral Array Coil Right VG	Right Lateral Coil Array
Lateral Array Coil Left VG	Left Lateral Coil Array
Medial Array Coil VG	Medial Coil Array
Lateral Single Loop Coil Left VG	Lateral Single Loop Coil Array
Medial Single Loop Coil Right VG	Medial Single Loop Coil Array
Lateral Single Loop Coil Right VG	Lateral Single Loop Coil Array
Medial Single Loop Coil Left VG	Medial Single Loop Coil Array

Coil Connections

Figure 2 demonstrates how to plug in the left lateral coil array into the connector. The same procedure is used for plugging in the medial coil array and right lateral coil array. Figure 3 demonstrates plugging in the medial plug when performing unilateral interventional procedures.

Figure 2: Plugging in the Left lateral Coil

Figure 3: Plugging in the Medial Plug

Environmental Requirements

The Vanguard should be stored at the same room temperature and relative humidity as your MR scanner.

Specifications

Coil System Specifications:

Field Strength: 1.5T

Frequency: 63.86MHz

Patient Table Specifications:

For use with GE Signa™ MRI Systems.

Maximum load capacity is 158.8kg or 350lbs.

The patient table is driven by the GE Signa™ MRI system.

Positioning accuracy, total patient support travel, and horizontal speed are controlled by the GE Signa™ MRI system and unchanged from the GE Signa™ patient table specifications.

The Vanguard cannot be raised or lower.

Coil Installation and Service

Coil Installation Procedure

Vanguard installation must be performed by a technician trained and approved by a Sentinelle Medical Inc. service representative.

Coil Service

The coils can not be serviced on site and must be returned to Sentinelle Medical Inc., address and contact information is available on page 34. The Vanguard can be serviced only by a technician trained and approved by Sentinelle Medical Inc.

Chapter
2

Chapter 2 Setup

General System Function and Components

The Vanguard is a dedicated breast imaging and interventional system for 1.5T GE Signa™ HDx configuration MR scanners. The Vanguard consists of a patient support with breast immobilization system, an attached 8 channel coil system and stretcher.

The Vanguard provides a large aperture which permits improved access to the patient's breasts during positioning. The patient support (Figure 4) holds the patient comfortably in a prone position with the breasts lying pendant into the aperture.

The immobilization system can operate in two different imaging modes; clinical and interventional, and consists of immobilization frames in sliders allowing for vertical and horizontal movement of the coils. The Vanguard's unique Variable Coil Geometry enables the coils to be positioned in close proximity to the breast tissue for improved signal-to-noise ratio (SNR).

For clinical imaging clear immobilization plates are placed in the immobilization frames to help minimize artifacts due to patient motion during scanning. For interventional procedures, these plates are replaced with sterile, single use disposable biopsy grids which serve to immobilize the breast and provide access windows for biopsy needle guidance.

Figure 4: Patient Lying Prone on Vanguard

System Components

The following describes the Vanguard's general set up for bilateral breast imaging, unilateral biopsy and bilateral biopsy procedures.

System Components for Bilateral Imaging Procedures

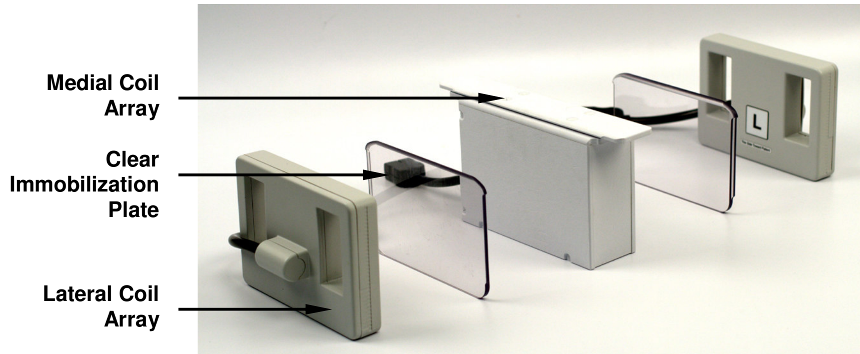


Figure 5: Bilateral Imaging Coils and Components

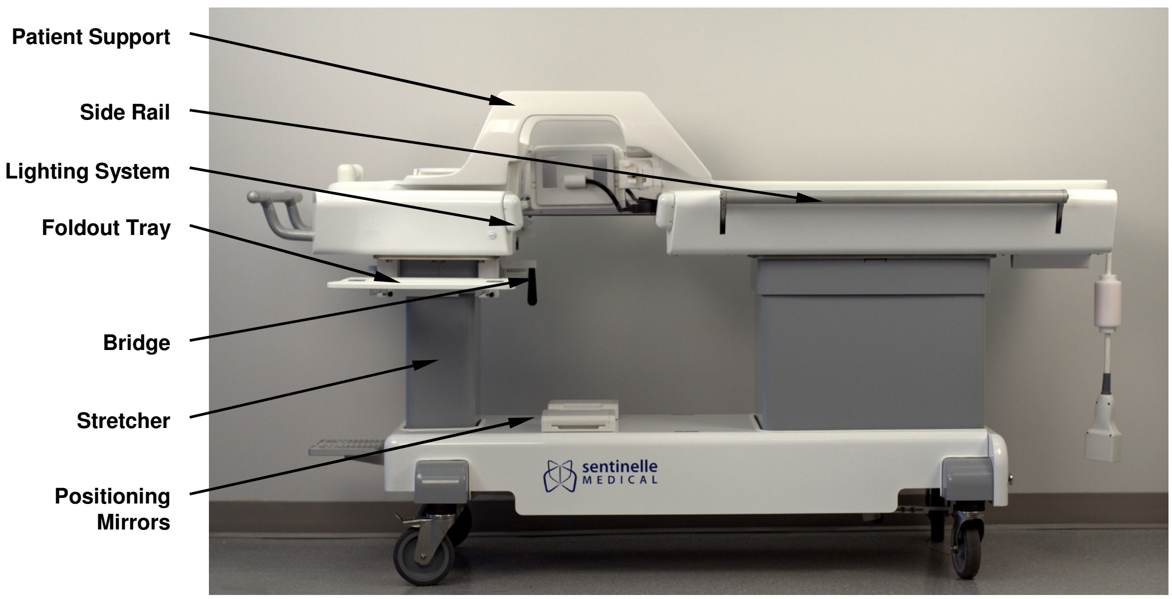


Figure 6: The Vanguard without Padding

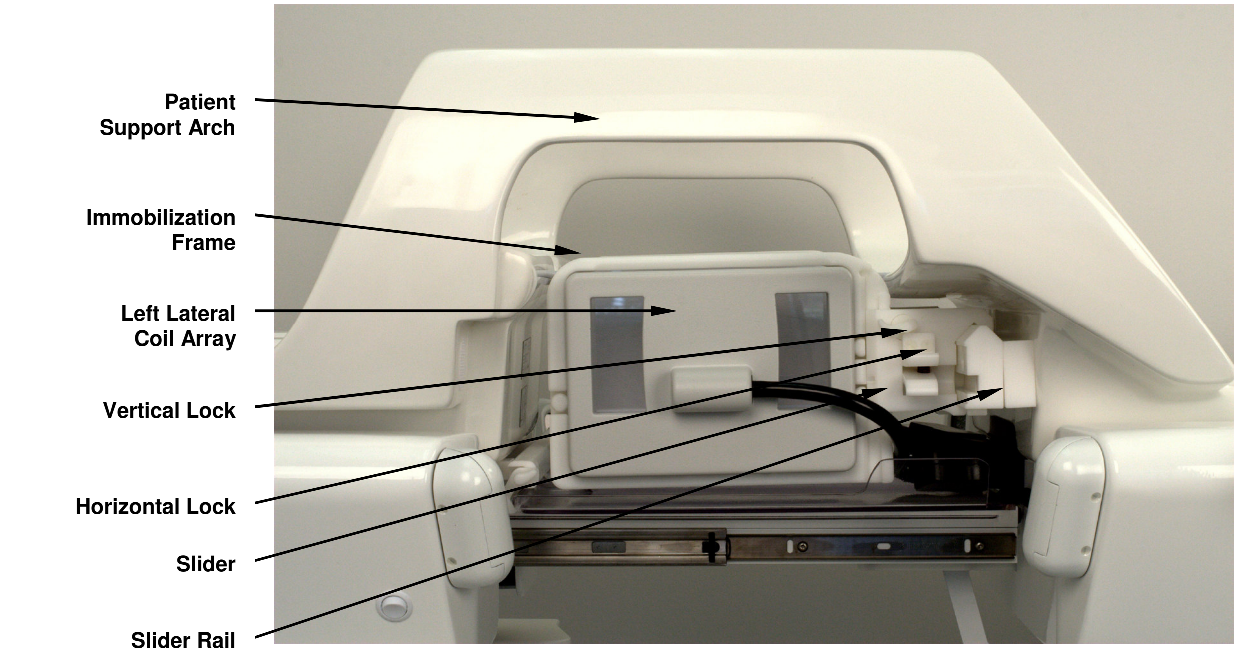


Figure 7: Vanguard Immobilization System

Figure 8 illustrates padding set-up on the Vanguard. Padding can be re-arranged for optimal patient comfort.

Figure 8: The Vanguard with Padding

System Components for Unilateral Interventional Procedures

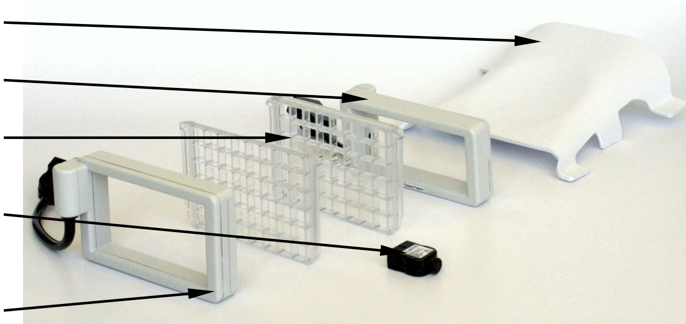


Figure 9: Coils and Components for Unilateral Interventional Procedures

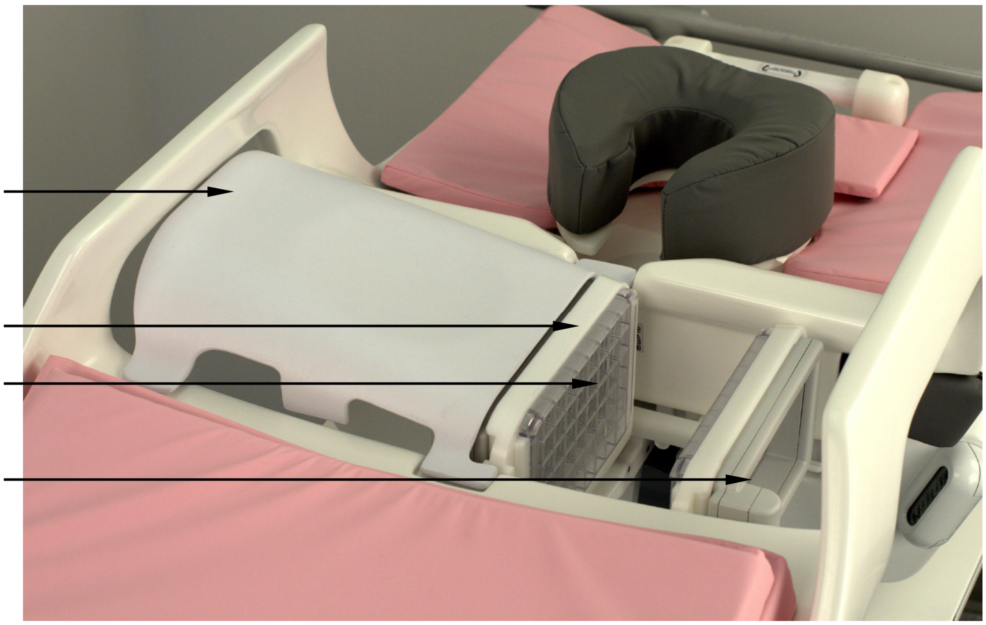


Figure 10: Set-up and Components for Unilateral Intervention Procedures

System Components for Bilateral Interventional Procedures

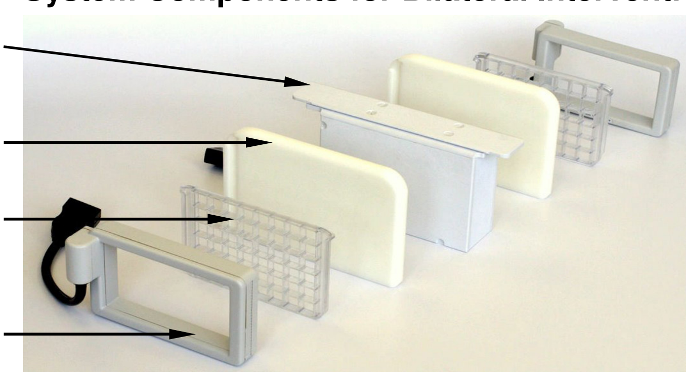


Figure 11: Coils and Components for Bilateral Interventional

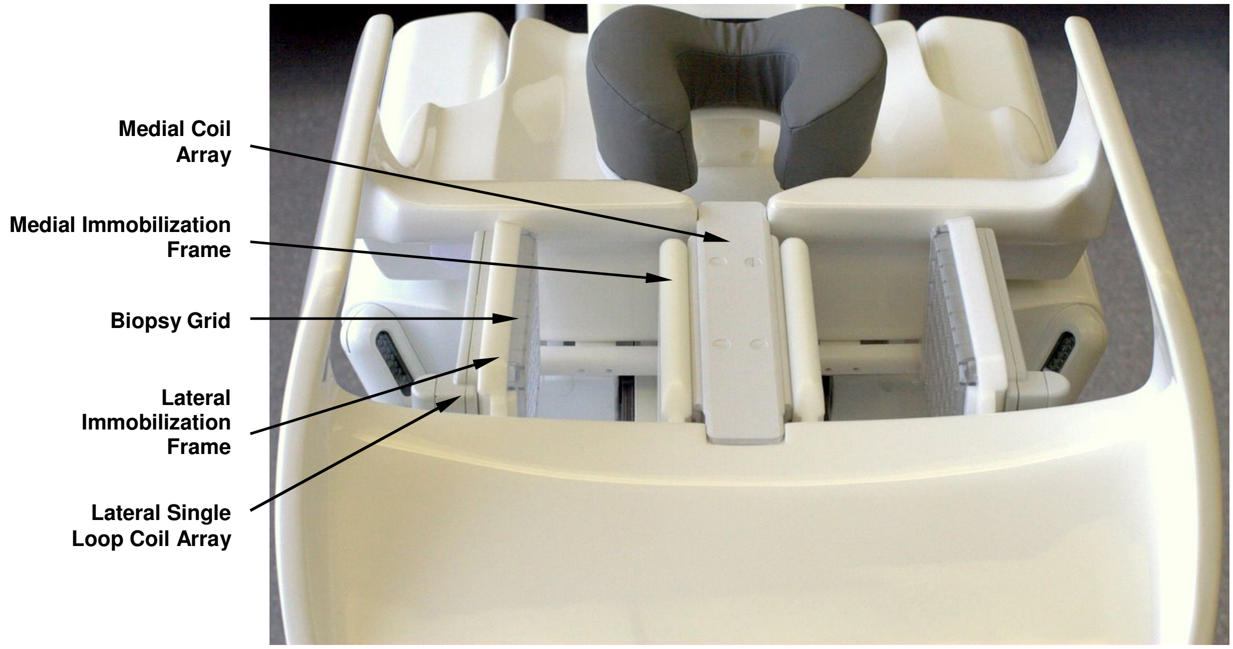


Figure 12: Set-up and Components for Bilateral Interventional Procedures

System Function

Vanguard Docking System

The Vanguard must be docked to the MR scanner using the docking pedals. The patient support will not advance to scan unless it is secured to the MR scanner. To dock and secure the Vanguard to the MR scanner, line up the patient foot end of the stretcher with the MR scanner, push the stretcher against the MR scanner and depress the **DOCK** foot pedal (Figure 13).

Undocking the Vanguard

With the patient support at the home position, the Vanguard can be undocked by depressing the **UNDOCK** pedal (Figure 13) and pulling the stretcher away from the MR scanner.

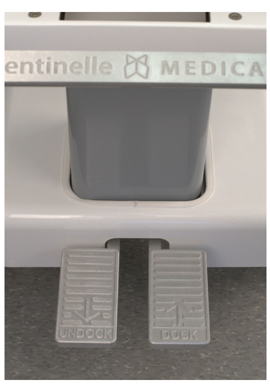


Figure 13: Docking Pedals on the Vanguard

Up/Down Adjustment of the Vanguard

Due to the unique geometry of the stretcher, users cannot adjust the height of the Vanguard.

Catchment Tray

Before advancing the patient, the patient support's catchment tray must be raised and latched and the bridge must be locked.

The Vanguard catchment tray lies horizontally between the immobilization system and the bridge. During interventional procedures, the catchment tray may be draped with an absorbent pad to catch any spills that may happen during the procedure. The catchment tray should be unlatched and opened to increase access to the breast during positioning. The catchment tray must be raised and latched into place before closing the bridge and advancing the patient (Figure 14).

Using the Bridge

The Vanguard has a retractable bridge which must be secured when advancing the patient support into and out of the MR scanner (Figure 15). When positioning a patient, the bridge can be retracted to permit the catchment tray to be opened, increasing access to the breast. To open the bridge, hold the bridge handle and slide it toward the head end of the patient support. To close the bridge, hold the bridge handle and slide it across the gap until the gap is completely closed and a clicking sound is heard, indicating the bridge is locked. **The bridge must be locked in order for the patient support to advance.**

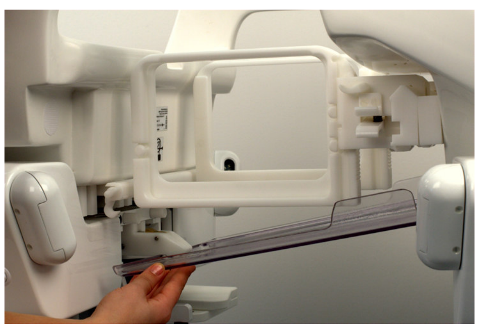


Figure 14: Raising or Lowering the Catchment Tray

Figure 15: Sliding the Bridge

Vanguard Cable

After docking the stretcher, plug in the Vanguard cable (Figure 16) to the *Connector A* port. Use caution to ensure that the Vanguard cable does not get caught between the patient support and the MR scanner when docking. A GREEN light should appear once the Vanguard cable is plugged in. Do not scan if the cable is unplugged.

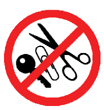
Figure 16: The Vanguard Cable

Side Rails

CAUTION

The Vanguard is equipped with one side rail on each side of the stretcher (Figure 6). To engage the side rails, pull the handle out and up to lock in place. Reverse this direction to stow the side rails in the stretcher. The side rail opposite to the side that the patient approaches the table should be in the upright position to help provide support while the patient mounts the Vanguard.

Drawers



WARNING

Two drawers (Figure 17) are located on the side of the Vanguard. These should only be used to store non-ferromagnetic Sentinelle Medical Inc. accessories such as biopsy grids, the contralateral breast support and coils.

Under no circumstance should there be any ferromagnetic material or non MRI compatible equipment stored in the drawers. Also, these drawers must not be used as a step under any circumstance.

Figure 17: The Vanguard Drawers

Lighting System

The Vanguard is equipped with lights on either side of the stretcher (Figure 6) to illuminate the region of interest when positioning the patient and during biopsy procedures. Light switches are located on both sides of the stretcher below the lights. There are two replaceable batteries located in the base of the stretcher. The battery life is approximately 10-12 hours. If the lights are dim or have changed to an orange colour, the batteries should be replaced. To access the batteries, loosen the screw and open the battery door (Figure 18). Loosen the screw on the plate that secures the batteries to the inside of the door and remove the plate to gain access to the batteries (Figure 19). Remove the two batteries and replace with two new ones (Figure 20).

CAUTION

The battery must be removed/replaced outside of the scan room. Once the batteries have been replaced, re-install the retaining plate by tightening the screw and secure the battery door with its screw.

Figure 18: Opening Battery Door

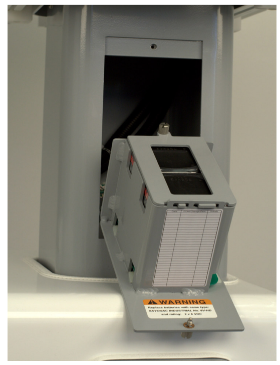


Figure 19: Accessing Battery

Figure 20: Replacing Batteries

Immobilization Frame Positioning

The immobilization frames can be adjusted both vertically and horizontally (medial/lateral and anterior/posterior). Each direction has its own locking mechanism. Depress the vertical adjustment lever down (Figure 21) to release the frame and slide it up or down. Once the frame is placed in the desired vertical position, lock it in place by releasing the lever. Pinch the horizontal lock (Figure 22) to release the frame slider and slide the frame left and right. Once the frame is placed in the desired position, lock it by releasing the lock.

The immobilization frames can be removed from the system entirely by depressing the vertical lock lever and sliding them straight up and out of their sliders. To remove the immobilization frames and sliders entirely from the slider rail, depress the slider rail button underneath and at the end of the slider rail and guide the sliders off the slider rail. To reinstall, reverse the process.

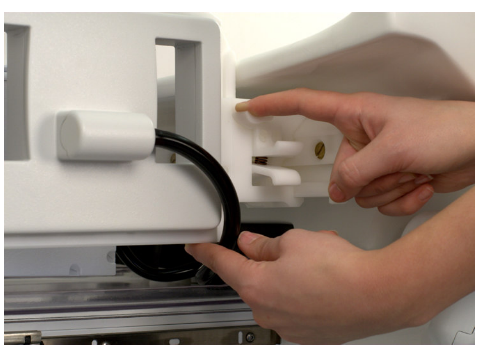


Figure 21: Adjusting Vertical Lock

Figure 22: Adjusting Horizontal Lock

Headrest

The headrest cushion may be separated from the headrest by pulling it off of its hook and loop attachment. This allows the headrest to be brought closer to the patient’s chin or further up towards the top of their head.

Adjust the height of the headrest by turning the knob located at the bottom of the headrest for optimal patient comfort, see Figure 23.

Figure 23: Adjusting Headrest Height

Bilateral Imaging with the Vanguard

The Vanguard is designed to allow patient preparation in a remote location outside of the MR suite, thus minimizing the time required in the MR Suite to perform a procedure. If desired, patient preparation can be performed inside the MR suite.

The following describes the general set up for Bilateral Breast Imaging. This procedure can be used for unilateral imaging as well.

Check List

- 2 Lateral Coil Arrays
- 1 Medial Coil Array
- 2 Clear Immobilization Plates
- Padding

The Vanguard allows the coils to be interchanged and reconfigured as appropriate for a particular imaging task; diagnostic bilateral imaging, unilateral biopsy and bilateral biopsy. Coils can be interchanged by first unplugging them, and pulling the coil straight out of the immobilization frame, storing them in the drawers. The desired coil is then pressed into the frame and plugged in. The plugs are keyed so that the coils can only be connected in the correct orientation.

The Vanguard coil design allows the user to position the coils in a customized position for each patient. This unique feature can accommodate various breast sizes and allows the coils to be positioned close to the breast for improved SNR. Any coils that are not in use must be stored in the Vanguard drawer.

Patient Positioning

1. Adjust the Vanguard padding to accommodate for patient size and comfort. Larger patients may require less padding.
2. Wipe down all padding with disinfectant and lay linens on the table.
3. Ensure the immobilization frames are open to the most lateral position.
4. Ensure the proper coils are attached and clear immobilization plates are inserted (Figure 24).

Figure 24: Inserting Clear Immobilization Plate

5. Ensure the side rail is up on the side opposite the side that the patient will approach the table.
6. Pull open the bridge and lower the catchment.
7. For all imaging, ensure the patient is wearing appropriate hearing protection.
8. Using a step stool, assist the patient onto the Vanguard ensuring they use the patient support arch for support as they lower their breasts into the aperture.
9. For patient comfort, lay the patient's arms by their side, ensuring there is no interference with the IV. Patients may also be positioned with arms resting above their head depending on patient size and comfort level.

WARNING

Ensure that the patient does not form a loop with any body parts. Do not allow patients to touch any part of the right hand or arm to any part of the left hand or arm. See Figure 25 and Figure 26 for incorrect and correct hand positioning, respectively. The loop formed by doing so could cause an RF burn at the point of contact. Use pads to ensure the patient's hands do not touch any metal features on the patient support. Be sure to educate the patient accordingly and check the patient's position before attempting to scan.

Figure 25: Incorrect Hand Positioning

Figure 26: Correct Hand Positioning

10. Stand at head end of table to ensure the patient is aligned centrally on the patient support. Have the patient adjust left/right position accordingly.
11. Approach the left side of the patient and turn on the Vanguard lighting system to improve visualization.
12. Have the patient move inferiorly or superiorly to ensure that the breast is centered in respect to the immobilization plate.
13. Adjust the headrest as needed at this time.
14. Pull the breast tissue down and away from the chest wall ensuring no tissue is caught on the posterior side of the immobilization plate or sternum support.
15. Using the mirrors (Figure 6), ensure that the breast is in profile.
16. Smooth breast tissue on the lateral side with thumb to remove mammary fold.

Note: The coil profile of the Sentinelle Medical Inc. lateral coil arrays have been optimized to allow sufficient signal posterior and anterior. Figure 27 shows the immobilization frames, which hold the lateral coils, set at optimal height. Typically, the frames should be 1 inch above the height of the medial coil array.

Figure 27: Immobilization Frame Set at Optimal Height

17. With coils set at optimal height, unlock the horizontal lock on the immobilization frames and move medially to offer immobilization but not compression of the breast. Lock frames.

18. Repeat positioning for right breast. Turn off the Vanguard lights.
19. Lift and latch the catchment, ensuring both locks are engaged and slide and lock the bridge. When an audible click is heard and the bridge remains in place the bridge is secured.
20. Place footrest pad under the patient's legs to improve comfort (Figure 8).
21. Ensure the Vanguard cable (Figure 16) is positioned on top of the patient support to prevent impingement during docking.

CAUTION

22. Do not touch the electrical pins in the connector housings (the system plug and the coil connectors in the cable tray, as well as the coils). Electrostatic discharge could damage the equipment.
23. Transport the patient into the MR suite, align the stretcher to the MR scanner and dock the Vanguard.
24. Ensure that the coil cables are protected by the catchment tray to ensure they do not become disconnected during patient support movement.

Figure 28: Patient Prone on the Vanguard

25. Advance the patient into the MR scanner, set landmark on the center of the coils and advance patient support to scan.
26. Proceed with imaging protocol.
27. At the end of the exam, shuttle the patient out of the MR scanner, transport the Vanguard out of the MR Suite, release immobilization of the breasts and assist the patient off the Vanguard.

Note: Please refer to Chapter 4 Maintenance/Troubleshooting in this manual for Vanguard cleaning instructions.

Interventional Imaging with the Vanguard

The following describes the general set up for a unilateral breast interventional procedure.

Check List

- 1 Contralateral Support
- 1 Lateral Single Loop Coil
- 1 Medial Single Loop Coil
- 2 Biopsy Grids
- 1-4 Fiducials
- 1 MRI Biopsy Worksheet
- Absorbent Pads
- Biopsy Supplies
- Post Operative Care Sheet
- Biohazard Waste Containers

Before starting any interventional procedure, it is important to review the patient's previous imaging. Knowing where the target is before the exam will assist in proper set up of the Vanguard and optimize patient positioning. Ideally, insert IV into opposite arm of the side of biopsy. This arm can be positioned above the patient's head for easy access during the procedure.

Attaching Frames and Coils

1. If in place, unplug and remove the medial coil array and lateral array coils. Store them in a drawer.
2. Unlock the horizontal locks of the immobilization frames and slide both frames to the side of biopsy (right for right breast and left for left breast).
3. Remove the clear immobilization plates from the immobilization frames and replace with biopsy grids, ensuring sterility is not compromised (Figure 29 and Figure 30).

Figure 29: Removing Clear Immobilization Plates

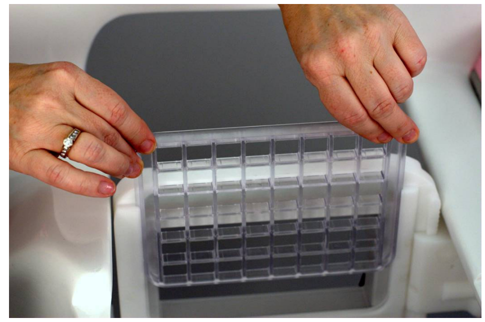


Figure 30: Inserting Biopsy Grid

4. Place the contralateral support to support the opposite breast during the procedure. Ensure the support is angled down lateral to medial. Position the medial immobilization frame beside and to the same height as the contralateral breast support. Lower slightly and lock in place.
5. Position the lateral immobilization frame to the most lateral position.
6. Click the lateral single loop coil and medial single loop coil into place ensuring that they are in the upper most position in the immobilization frames and that the cables are on the side toward the patient's feet.

Note: The lateral single loop coil is larger than the medial single loop coil.

7. Plug the cables from the coils into the lateral connectors attached to the table. Ensure the medial plug is inserted into the medial connector.

Contralateral Breast Support

The contralateral breast support is designed to support the contralateral breast away to improve medial access during unilateral interventional procedures. The design of the breast support puts the patient in a slightly oblique position to allow the breast of interest to fall further into the open aperture for increased posterior access to the breast during biopsy.

Patient Preparation

1. Adjust the Vanguard padding to accommodate for patient size and comfort. Larger patients may require less padding.
2. Wipe down all padding with disinfectant and lay linens on the table.
3. Pull open the bridge and lower the catchment tray.

Note: To provide additional immobilization, you may use a sterile covering between the patient and the sterile biopsy grid. Placing a sheet of OPSITE™ Flexifit (Smith and Nephew part # 66000375) with the adhesive side against the breast and the smooth side towards the grid serves as a barrier, and also enables immobilization of the breast in a more uniform manner. Ensure the entire surface contacting the patient is covered. OPSITE™ Flexifit is a commercially available sterile barrier used for wound dressing.

4. For all imaging, ensure the patient is wearing appropriate hearing protection.

5. Clean and prepare the patient’s breast according to site policies and procedures.
6. Using a step stool, assist the patient onto the Vanguard ensuring they use the patient support arch as they lower their breasts into position. The target breast will be between the two biopsy grids and the opposite breast will rest comfortably on the contralateral breast support.
7. Stand at head end of table to ensure the patient is aligned centrally on the patient support. Have the patient adjust left/right position accordingly.
8. Turn on the Vanguard lighting system to improve visualization during positioning.
9. Have the patient move inferiorly or superiorly to ensure that the breast is centered in respect to the biopsy grid.
10. Adjust the headrest as needed for optimal patient comfort.
11. Pull the breast tissue down and away from chest wall. Apply light immobilization by first moving the medial frame. Once positioned, ensure the medial frame is locked both horizontally and vertically.
12. Apply sufficient immobilization from the lateral frame to keep the breast in position. The breast should feel like the palm of your hand when fingers are extended (Figure 31).
13. Place fiducial(s) (provided by Sentinelle Medical Inc.) or vitamin E capsule in a grid window in close proximity to the target. Document this location on the appropriate Grid Worksheet (Figure 32).

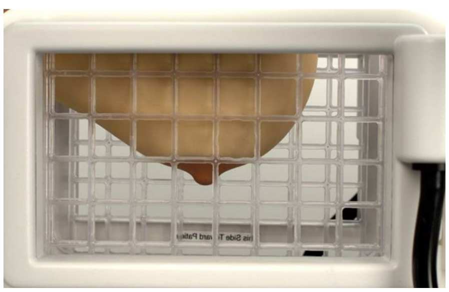


Figure 31: Breast Positioned in Grid

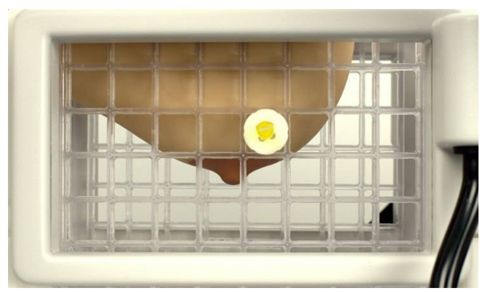


Figure 32: Fiducial Marker in Grid

14. Lift and latch the catchment, ensuring both locks are engaged and slide and secure the bridge.
15. Place footrest pad under the patient’s legs to improve comfort.
16. Ensure the Vanguard cable is positioned on top of the patient support to prevent impingement during docking.

CAUTION

17. Do not touch the electrical pins in the connector housings (the system plug and the coil connectors in the cable tray, as well as the coils). Electrostatic discharge could damage the equipment.

18. Transport the patient into the MR suite, align the stretcher to the MR scanner and dock the Vanguard.
19. Ensure that the coil cables are protected by the catchment tray to ensure they do not become disconnected during patient support movement
20. Advance the patient into the MR scanner, set landmark on the center of the coils and advance to scan. Ensure that patient or Vanguard does not impede the transport of the patient support.
21. Proceed with imaging protocol.
22. At the end of the exam, shuttle the patient out of the MR scanner, undock the Vanguard from the scanner and transport the Vanguard out of MR Suite. Release immobilization of the breasts and assist the patient off the Vanguard.
23. Unlock the immobilization frame and remove from slider. There is a release button on the under-side of the slider which must be depressed in order to remove the frame.
24. Apply necessary compression, steri-strips, gauze and bandage to the biopsy site as per the site's recommendations.
25. After patient has been assisted off the Vanguard, wipe down with appropriate disinfectant. Dispose of biopsy grids in biohazard container as per institution practice.

Note: Please refer to Chapter 4 Maintenance/Troubleshooting in this manual for Vanguard cleaning instructions.

The following describes the general set up for bilateral interventional procedures.

Check List

- 2 Lateral Single Loop Coils
- 2 Medial Immobilization Frames
- 1 Medial Coil
- 2 Biopsy Grids
- 2-6 Fiducials
- 1 MRI Biopsy Worksheet
- Absorbent Pads
- Biopsy Supplies
- Post Operative Care Sheet
- Biohazard Waste Container

Before starting any interventional procedure, it is important to review the patient's previous imaging. Knowing where the targets are before the exam will assist in proper set up of the Vanguard and optimize patient positioning.

Attaching Frames and Coils

1. If in place, unplug and remove the lateral array coils. Store them in a drawer.
2. Additional immobilization frames are required for bilateral interventional procedures (Figure 11). Remove the lateral immobilization frames from the slider by pinching the horizontal lock and sliding the frames laterally for removal. Insert the medial immobilization frames first, followed by the lateral immobilization frames.
3. Remove the clear immobilization plates from the lateral immobilization frames and replace with biopsy grids (Figure 29 and Figure 30).
4. Ensure that the medial coil array is in place. Insert the lateral single loop coils in the immobilization frames, ensuring that they are in the upper most position and that the cables are on the side toward the patient's feet. Position the lateral immobilization frames to the most lateral position (Figure 12).
5. Insert the coil cables into the appropriate connectors of the Vanguard.

Patient Preparation

1. Adjust the Vanguard padding to accommodate for patient size and comfort. Larger patients may require less padding.
2. Wipe down all padding with disinfectant and lay linens and absorbent pads on the Vanguard as per normal practice.
3. Retract the bridge and lower the catchment tray.

Note: To provide additional immobilization, you may use a sterile covering between the patient and the sterile biopsy grid. Placing a sheet of OPSITE™ Flexifit (Smith and Nephew part # 66000375) with the adhesive side against the breast, and the smooth side towards the grid serves as a barrier, and also enables immobilization of the breast in a more uniform manner. Ensure the entire surface contacting the patient is covered. OPSITE™ Flexifit is a commercially available sterile barrier used for wound dressing

4. For all imaging, ensure the patient is wearing appropriate hearing protection.
5. Prepare the patient's breasts according to site policies and procedures (cleansing with disinfectant, Vitamin capsules, etc).
6. Using a step stool, assist the patient onto the Vanguard ensuring they use the patient support arch for support as they lower their breasts into position.
7. Stand at head end of table to ensure the patient is aligned centrally on the patient support. Have the patient adjust left/right position accordingly.
8. Turn on the Vanguard lighting system to improve visualization during positioning.
9. Have the patient move inferiorly or superiorly to ensure that the breast is centered in respect to the biopsy grids.
10. Adjust the headrest using the knob as needed at this time, and position the head rest pad as required for optimal patient comfort.
11. Pull the breast tissue down and away from chest wall. Apply light immobilization by first moving the medial frame. Once positioned, ensure the medial frame is locked both horizontally and vertically.
12. Apply sufficient immobilization from the lateral biopsy grid to keep the breast in position. The breast should feel like the palm of your hand when fingers are extended (Figure 31).
13. Place fiducial(s) (provided by Sentinelle Medical Inc.) or a vitamin E capsule in a grid window in close proximity to the target. Document this location on the appropriate Grid Worksheet. Repeat steps 11 to 13 for the other breast (Figure 32).
14. Lift and latch the catchment, ensuring both locks are engaged and slide and secure the bridge.
15. Place footrest pad under the patient's legs to improve comfort.
16. Ensure the Vanguard cable (at the foot end of the table) is positioned on top of the stretcher as to not get impinged during docking.
17. Ensure that the coil cables are protected in the catchment tray to prevent disconnection during patient support movement.

CAUTION

18. Shuttle the patient into the MR scanner, landmark on the center of the coils. Ensure that patient or Vanguard components do not impede the transport of the patient support.

19. Proceed with imaging protocol.
20. At the end of the exam, shuttle the patient out of the MR scanner, undock the Vanguard from the scanner and transport the Vanguard out of the MR Suite.
21. Unlock the immobilization frame and pull off from slider. There is release button on the under-side of the slider which must be depressed in order to remove the frame.
22. Apply necessary compression, steri-strips, gauze and bandage to the biopsy site as per the site's recommendations.
23. After patient has been assisted off the Vanguard, wipe down with appropriate disinfectant. Dispose of biopsy grids in biohazard container as per institution practice.

Note: Please refer to Chapter 4 Maintenance/Troubleshooting in this manual for Vanguard cleaning instructions.

Recommended Scan Protocols

Any specific choice of protocols is entirely the responsibility of the radiologist and/or technologist. A set of protocols commonly used by breast MR sites can be provided by contacting Sentinelle Medical Inc.

Chapter
3

Chapter 3 Quality Assurance (QA)

Instructions for MR Imaging on Phantoms

Sentinelle Medical recommends that the coils be checked for imaging integrity on a regular basis, by performing the Quality Assurance (QA) procedure. The QA procedure involves a phantom scan, a visual inspection of the QA images and a calculation of the signal-to-noise ratio. The phantoms used for the QA scan will be provided by a Sentinelle Medical Service representative. Only use these phantoms when performing QA scans on the Vanguard.

Note: If bubbles are apparent on the in the phantom, tap the outside of the phantom until these bubbles disappear.

Test 1: 8-Channel Bilateral Breast SNR Verification

1. Dock Vanguard to the MR scanner and insert the Vanguard cable into the scanner.
2. Set up coils for a bilateral breast exam. For optimal imaging quality, both clear immobilization plates shall be positioned at the same height as each other. Position the top of the immobilization frames 1 inch higher than the medial coil.
3. Place the Vanguard phantoms between the immobilization frame and medial coil array. Ensure the Vanguard catchment tray and bridge is closed.
4. At the MR console, prescribe the following scan.

- Coil configuration: click the tab next to **Coil**, and click **Currently Connected**. Then select: **8ch Bilat HD Breast Array by Sentinelle** then **8 BrstBL SMI.**
- Description: **8Ch SNR**

Imaging Parameters:

Scan Plane:	Axial	
	Field of View:	32
	Pulse Sequence:	SE – single echo
	TR:	450
	TE:	14
	Start Location:	S0.0
	End Location	I0.0
	# Slices:	1
	Slice Thickness:	5
	Spacing	0
	NEX:	1
	Matrix Size:	256x256
	BW	15

5. Display the image using the image browser on the MR console. Select a rectangular Region of Interest (ROI) positioned entirely within the left phantom and encompassing about 75% of the signal-producing region of the phantom.
6. Record the mean value for this ROI in the **Left Signal Mean** section of Table 4-5.
7. Select another rectangular ROI, the same size as the first, and position it entirely within the signal producing region of the right phantom. Record the mean value for this ROI in the **Right Signal Mean** section of the Quality Assurance SNR Table below. Some useful shortcuts when working with ROIs: **Crtl-C** (to copy), **Crtl-V** (to paste), **Crtl-X** (to delete).
8. Finally select a third ROI in a region outside of both phantom signals in the image. Ensure that this ROI is at least 2 cm across. Record the standard deviation for this ROI in the **Noise Standard Deviation** section of the Quality Assurance SNR Table below.
9. Calculate the Signal-to-Noise Ratio (SNR) and record it in the Quality Assurance SNR Table. SNR is computed using the following equation:

$$\mathbf{SNR} = \frac{\textbf{(Left Signal Mean + Right Signal Mean) / 2}}{\textbf{Noise Standard Deviation}}$$

Note: All values should be entered into Table 4-5 and compared with baseline SNR measurements (values from tests performed during installation). Acceptable SNR values should have a variance of ±25% of the baseline values. Larger or smaller value may be indicative of problems with the coils or imaging system. Please contact Sentinelle Medical Inc. at 1-866-735-3744 if these values are not within an acceptable range.

Chapter
4

Chapter 4 Maintenance/Troubleshooting

System Care

Servicing testing of the table docking and advancing of the patient support into the MR scanner must be performed by a qualified Sentinelle Medical Inc. Service Representative.

Storage

The Vanguard must be stored at the same room temperature and relative humidity as the MR system.

Transport and store the Vanguard and all components under the following environmental conditions only, for a maximum of four weeks.

Table 4-1: Transportation and Storage Environment (4 week maximum)

	Minimum	Maximum
Ambient Temperature (°C)	+10	+40
Relative Humidity (non-condensing)	10%	90%
Atmospheric Pressure	500 Hg	1060 Hg

Inspection

CAUTION

The coil and mechanical apparatus should be inspected weekly for mechanical damage or breakage. Inspect to see if system is docking securely and table motion into magnet is smooth and regular. DO NOT USE THE SYSTEM IF IT HAS SUSTAINED MECHANICAL DAMAGE OR IF TABLE MOTION IS SIGNIFICANTLY COMPROMISED. Contact a Sentinelle Medical representative immediately.

It is important to inspect the coils to ensure the enclosures are intact. If the coil enclosure appears to be compromised, contact a Sentinelle Medical Service Representative. A replacement coil can be purchased unless under warranty.

Special Care Requirements

Cleaning

The Sentinelle Medical biopsy grids are sterile, single use disposable items. The following parts, however, should be cleaned prior to and after each use.

Note: Protective gloves are required when performing the procedures set forth in these instructions, as well as compliance with your site's biohazard/blood-borne pathogen safety protocols.

Routine Imaging:

- The Vanguard Immobilization System
- Padding

Interventional Imaging:

- The Vanguard Immobilization System
- Padding
- Fiducials
- Single Loop Coil Arrays
- Medial Coil Array (for bilateral interventional procedures)

The cleaning solutions below have been tested and are recommended for cleaning certain components of the Vanguard. Use a cotton cloth to clean the coil, surfaces of the system, and padding. Follow cleaning instructions below accordingly.

CAUTION

The coils are not waterproof and should not be immersed in any liquid. This will damage the coils and may cause injury.

Table 4-2: Approved Cleaning Solutions

Solution	Applicable Use
Warm water	Safe for all areas
Commercial dishwashing liquid/water combination	Safe for all areas
Alcohol solution (70%, isopropyl alcohol, 30% water).	Do not apply to adhesive-backed materials such as labels and Velcro® fasteners.
Virox disinfectant	Do not apply to adhesive-backed materials such as labels and Velcro® fasteners.
Anticeptizyme	Use for immobilization frames, patient support and coils as needed to break up bodily fluids.

NOTES:

- Do not apply tape to top surface of padding.
- Do not mark padding with ink.
- Do not immerse coil or coil cable in cleaning fluid.
- If bodily fluids contact immobilization frames, coil enclosures, or catchment tray of patient support body, use appropriate decontamination techniques to clean.
- Ensure coil enclosure is intact.
- Coils cannot be sterilized using autoclaving techniques.

Replaceable Components/Accessories

Table 4-3 lists the system accessories that may be purchased for replacement:

Table 4-3

Part Name	Sentinelle Part Number
Biopsy Grids (Box of 10)	5000137-11
Padding Set - Full	4000021-11
Headrest Pad	5000065-11
Immobilization Plate	5000140-11
Batteries	6V-HDM

Troubleshooting

Please contact a Sentinelle Medical Inc. representative between the hours of 7AM and 7PM Monday to Friday (North American Eastern Standard Time) to arrange for service or repair of the system. Also describe your problem in an email addressed to feedback@sentinellemedical.com

Correspondence can be sent to:

Customer Service
Sentinelle Medical Inc.
3080 Yonge St. Suite 6020.
Toronto, Ontario, Canada
M4N 3N1

Tel: 866-735-3744
Fax: 416-482-3807

Appendix 1

Table 4-4: Baseline SNR measurements

Date	Left Signal Mean	Right Signal Mean	Noise Standard Deviation	SNR

Table 4-5: Quality Assurance SNR

Date	Left Signal Mean	Right Signal Mean	Noise Standard Deviation	SNR